Lab Report – EE-170L Computer Programming (Lab 9)

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Objectives

- Understand the concept of functions in C++.
- Learn how to declare, define, and call functions.
- Practice writing functions with and without parameters.
- Apply functions to solve problems using arrays.

📖 Theory

Functions in C++ allow us to break programs into smaller reusable parts. Each function has:

- A **name** (identifier).
- **Parameters** (optional inputs).
- A **return type** (the kind of value it gives back).

The **prototype** tells the compiler what the function looks like before it is used.

Syntax:

```
returnType functionName(parameters);
```

Examples:

- int main(); → returns an integer, no parameters.
- float divide(int a, int b); → returns a float, takes two integers.
- void square(float x); → returns nothing, takes one float.

Lab Tasks

Task 1: Function Introduction and Declaration

```
#include <iostream>        // include input/output library

using namespace std;        // use standard namespace
```

```
void greet() {            // function declaration with no parameters
   cout << "Welcome to Lab 09!"; // print greeting message
}

int main() {              // main function starts
   greet();               // call the greet function
   return 0;              // end program
}
```

Task 2: Basic Function Declaration and Call

```
#include <iostream>       // include input/output library
using namespace std;      // use standard namespace

void sayHello() {         // function declaration with no parameters
   cout << "Hello, World!"; // print hello message
}

int main() {              // main function starts
   sayHello();            // call the sayHello function
   return 0;              // end program
}
```

Task 3: Sum and Average of Array Elements

```cpp
#include <iostream>               // include input/output library
using namespace std;              // use standard namespace

int calculateSum(int arr[], int size) { // function to calculate sum
   int sum = 0;                           // variable to store sum
   for(int i = 0; i < size; i++) {        // loop through array
      sum += arr[i];                      // add each element to sum
   }
   return sum;                            // return total sum
}

int main() {                              // main function starts
   int numbers[10];                       // declare array of size 10
   cout << "Enter 10 integers: ";         // ask user for input
   for(int i = 0; i < 10; i++) {          // loop for 10 inputs
      cin >> numbers[i];                  // store each input in array
   }

   int total = calculateSum(numbers, 10); // call function to get sum
   float average = total / 10.0;          // calculate average

   cout << "Sum = " << total << endl;             // display sum
   cout << "Average = " << average << endl; // display average
```

```
    return 0;                                   // end program
}
```

Results

- Task 1 displayed: Welcome to Lab 09!
- Task 2 displayed: Hello, World!
- Task 3 correctly calculated the sum and average of 10 integers entered by the user.

Conclusion

In this lab, I learned how to:

- Declare and define functions in C++.
- Call functions from main().
- Pass parameters to functions and return values.
- Use functions to simplify array operations like sum and average.

Functions make programs **modular, reusable, and easier to understand**.